

**Cross-Cultural Buddy System: Validating Interest-Based Peer Matching as a Pathway to
Sociocultural Adaptation Among Chinese International Students in UK Universities**

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Abstract

Chinese international students in UK universities face significant sociocultural adaptation challenges, characterised by ethnic clustering, unmet psychological needs, and limited cross-cultural contact. This study proposes a longitudinal quasi-experimental validation of the Cross-Cultural Buddy System, an interest-based peer-matching intervention pairing two Chinese international students with one UK home student. Grounded in Homophily (McPherson et al., 2001), Self-Determination Theory (Ryan & Deci, 2000), and Intergroup Contact Theory (Allport, 1954), the study examines whether interest-based matching yields greater improvements in SDT need satisfaction and sociocultural adaptation compared to eligible non-participants. Data will be collected at three time points across the first academic year using the Basic Psychological Needs Scale and the Sociocultural Adaptation Scale. It is hypothesised that buddy system participants will show significant improvement in SDT need satisfaction and need fulfilment will mediate improvements in sociocultural adaptation outcomes.

Keywords: sociocultural adaptation, self-determination theory, intergroup contact, homophily, Chinese international students

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Chinese international students represent one of the largest international student groups in UK universities, with a 56% growth over the last five years (Ploner, 2018). However, large numbers do not equate to smoother integration. Unlike domestic students, Chinese international students face a double transition, adapting to both academic and cultural shifts simultaneously, while navigating language barriers, cultural adjustment difficulties, and limited social integration (Holliman et al., 2024). This dual burden challenges their sociocultural adaptation, defined as the behavioural and cognitive adjustments required to function effectively within a new cultural environment (Ward & Kennedy, 1999), encompassing skills such as navigating social interactions and understanding local norms. English fluency anxiety specifically undermines perceived competence, making cross-cultural interaction feel threatening rather than rewarding. Meanwhile, limited familiarity with UK social norms reduces opportunities for meaningful connection, threatening relatedness. Co-national clustering, while a rational short-term coping response, inadvertently reinforces social distance from host students and restricts long-term sociocultural adaptation (Ploner, 2018). Existing interventions such as welcome week events tend to be high-pressure and transient, lacking the progressive structure needed to sustain cross-cultural engagement (Zhu & Peng, 2026). It is against this background that the following theoretical framework, integrating Homophily, Self-Determination Theory and Intergroup Contact Theory, is proposed to explain both the psychological mechanisms underlying this transition challenge and the design logic of the buddy system intervention.

Homophily refers to the tendency of individuals to associate with similar others. This pattern is observed across multiple dimensions of differentiation, including race, gender, socioeconomic

status, and shared interests (McPherson et al., 2001). By interacting predominantly with similar others, individuals reinforce shared experiences and perspectives, coming to typify what McPherson et al. (2001) describe as "people like us." This pattern is so deeply rooted in social behaviour that Plato observed in Phaedrus that "similarity begets friendship" (Plato, as cited in McPherson et al., 2001). Empirically, as similarity between interaction partners increases, so does liking, cooperation, and trust (Clerke & Heerey, 2021), creating the psychological conditions necessary for SDT's competence and relatedness needs to be fulfilled.

Self-Determination Theory (SDT; Ryan & Deci, 2000) proposes that three basic psychological needs, autonomy, competence, and relatedness, are essential for facilitating growth, integration, and personal well-being. For Chinese international students transitioning to UK universities, these needs are particularly threatened. Limited intercultural interaction and low perceived social support undermine relatedness, while ethnic clustering further restricts opportunities for cross-cultural connection (Ploner, 2018). The cross-cultural buddy system operationalises SDT by pairing Chinese international students with UK home students based on shared interests (2:1 ratio), providing an immediate foundation for relatedness and conversational competence, while voluntary participation beyond structured meetups preserves autonomy. Critically, if SDT need satisfaction increases without corresponding improvements in sociocultural adaptation, this would challenge the assumption that need fulfilment is the operative mechanism, rather than social exposure alone.

Intergroup Contact Theory, originally proposed by Allport (1954), posits that direct contact between members of different social groups reduces intergroup prejudice and social distance if contact occurs under conditions of institutional support, equal status, cooperative goals, and opportunities for personal acquaintance. The buddy system addresses this through institutional

embedding within the university structure, pairing Chinese students with UK home students in a low-pressure, repeated-contact format. While true equal status may be difficult to achieve in given cultural familiarity asymmetries, the 2:1 ratio partially addresses this by ensuring no single Chinese student bears the full communicative burden, while also guarding against over-reliance on a single host student (Pekerti et al., 2021). Furthermore, the presence of two Chinese students within each triad provides mutual linguistic and emotional support, reducing language anxiety and lowering the threshold for meaningful participation in cross-cultural exchange. Crucially, host-national friendships significantly predict both psychological and socio-cultural adaptation outcomes, whereas co-national friendships do not demonstrate this moderating effect (Sheng et al., 2022). However, if structured contact increases cross-cultural interaction without altering SDT need satisfaction or adaptation outcomes, this would suggest that contact alone is insufficient and the quality of interaction, mediated by Homophily and need fulfilment is the operative mechanism rather than contact frequency per se.

While buddy schemes have been implemented across UK universities (Lesk & Montaldo, 2018), the relationship between structured peer contact and psychological well-being among Chinese international students remains underexplored, and to the author's knowledge, no validated intervention has integrated interest-based matching with SDT need fulfilment within a structured contact framework for this population. This study therefore proposes a longitudinal quasi-experimental validation of the Cross-Cultural Buddy System, examining whether interest-based peer matching produces greater improvements in SDT need satisfaction and sociocultural adaptation compared to eligible non-participants. Data will be collected at three time points across the first academic year, starting with pre-arrival (T1), end of first term (T2), and end of first year (T3), using self-report measures of SDT need satisfaction and sociocultural adaptation.

It is hypothesised that: (H1) buddy system participants will show significantly greater increases in SDT need satisfaction, particularly relatedness and competence, compared to non-participants across the academic year, and (H2) greater need satisfaction will significantly predict improvements in sociocultural adaptation, such that need fulfilment mediates the relationship between programme participation and adaptation outcomes. A mixed ANOVA will be used to examine Group X Time interaction across time points, followed by mediation analysis for H2 testing. Lastly, this validation study is situated within the Validate stage of the Design Psychology Model (van Zalk & Sheldrick, 2025), a framework for integrating psychological science into design practice through iterative cycles of grounding, validation, and monitoring of interventions.

Method

Participants

Participants will be first year Chinese international undergraduate and postgraduate students enrolled at a UK university, alongside eligible UK home students volunteering as buddy partners. Recruitments will be placed through the university's international student office prior to arrival. Additionally, UK home students will be recruited through university notice boards and student union communications, with entirely voluntary participation. Students who opted out of the programme but consented to survey participation will form the comparison group, allowing for quasi-experimental comparison of outcomes across the year. Comparison group participant will be identified through opt-out records maintained by the international student office, ensuring comparable eligibility criteria across both groups. A priori power analysis was conducted using G*Power (Faul et al., 2007), with parameters set as a medium effect size ($f = 0.25$; Cohen, 1988), consistent with comparable cross-cultural intervention studies (Szabó et al., 2020), α

= .05, and 80% power, yielding a minimum sample size of 52 participants per group. Accounting for an anticipated 20% attrition rate across three longitudinal time points, a target sample of 64 participants per group (128 total) was established. Eligible participants must be enrolled full-time in their first year, with no prior experience of studying in the UK. All participation will be voluntary, with informed consent obtained prior to data collection and the right to withdraw at any time without penalty. Data will be anonymised using coded participant identifiers in accordance with GDPR guidelines and the British Psychological Society's Code of Ethics and Conduct (2021).

Design

A longitudinal quasi-experimental study will be conducted, with data collected at three time points: pre-arrival (T1), end of first term (T2), and end of first year (T3). A longitudinal design was selected as sociocultural adaptation is a gradual process unlikely to be fully captured within a single time point (Szabó et al., 2020). The independent variable is programme participation, with two levels: buddy system participants (intervention group) and eligible students who opted out (comparison group). Participants are allocated to conditions through self-selection (an independent groups design) reflecting the voluntary nature of the programme. The dependent variables are SDT need satisfaction and sociocultural adaptation, measured at each time point.

Materials

SDT Need Satisfaction will be measured using the Basic Psychological Need Scale (BPNS; Gagné, 2003), a validated self-report measure assessing autonomy, competence, and relatedness on a seven-point Likert scale. This scale was selected because it directly operationalises the three needs central to the theoretical framework and has demonstrated strong internal consistency in

Chinese student populations (Cronbach's $\alpha = 0.80-.89$; Gagné, 2003), making it particularly appropriate for the present sample.

Sociocultural Adaptation will be measured using the Sociocultural Adaptation Scale (SCAS; Ward & Kennedy, 1999), a 29-item self-report measure assessing the degree of difficulty experienced in navigating the host culture across social, academic, and everyday domains. The SCAS was selected because it specifically captures behavioural and cognitive adjustment to a new cultural environment, rather than general well-being, making it the most appropriate measure of the adaptation outcome targeted by this intervention. The scale has demonstrated robust internal consistency across culturally diverse student samples ($\alpha = .75-.91$; Ward & Kennedy, 1999).

Procedure

Prior to arrival, all incoming Chinese international students will be invited to complete an online interest-matching form capturing course, language spoken, hobbies, and age range. An algorithm will pair two Chinese international students with one UK home student based on weighted similarity scores, prioritising course alignment, shared interests, and language overlap, while minimising nationality clustering. Matched triads will be introduced via a WhatsApp group before the academic year begins.

During orientation week, all participating triads will attend a university-organised lunch event, representing the first structured face-to-face meeting. From week two onwards, the triads will be encouraged to meet voluntarily for weekly lunchtime meetups, incentivised through a £3 meal discount per person.

Surveys measuring SDT need satisfaction and sociocultural adaptation will be administered online at T1(pre-arrival), T2 (end of first term, tentatively week 10), and T3 (end of first year,

tentatively week 30). Comparison group participants, eligible students who opted out of the programme, will complete identical surveys at the same time points without any programme involvement. These surveys will be distributed via a secure online platform (e.g., Qualtrics, Microsoft Forms) and completed independently by participants. A mixed ANOVA will be conducted to examine Group $\times$ Time interactions in BPNS and SCAS scores across the three time points, with a subsequent mediation analysis testing whether changes in SDT need satisfaction mediate the relationship between programme participation and sociocultural adaptation outcomes (Hayes, 2018).

Ethics

This study will adhere to the British Psychological Society's Code of Ethics and Conduct (2021). All participants will provide informed consent prior to participation and retain the right to withdraw at any time. Data will be anonymised using coded participant IDs to ensure confidentiality, and all data will be stored securely in accordance with GDPR guidelines. As the study involves no deception or vulnerable populations beyond the general student context, standard ethical approval procedures will apply. The Ethical approval will be obtained from the university's institutional ethics committee prior to data collection.

Results

The hypothesis for this study were based on examining whether interest-based peer matching produces greater improvements in SDT need satisfaction and sociocultural adaptation compared to eligible non-participants. It is predicted that a significant Group $\times$ Time interaction will emerge from the mixed ANOVA, indicating that buddy system participants will show greater increases in BPNS scores, particularly relatedness and competence subscales, compared to non-participants across the three time points (T1, T2, T3), thereby supporting H1. Effect sizes will be

reported using partial eta squared (η^2p), with a medium effect ($\eta^2p = .06$) anticipated, consistent with comparable cross-cultural intervention studies (Szabó et al., 2020).

Regarding H2, mediation analysis is expected to reveal that changes in SDT need satisfaction significantly mediate the relationship between programme participation and sociocultural adaptation outcomes, as measured by the SCAS. The indirect effect of programme participation on SCAS scores through BPNS change is predicted to be significant, consistent with SDT's proposition that need fulfilment rather than social exposure alone drives sustained integration (Ryan & Deci, 2000). The indirect effect will be estimated using bootstrapping with 5,000 resamples, with 95% confidence intervals reported (Hayes, 2018).

Conversely, several null results would carry important theoretical implications. If no significant Group $\times$ Time interaction emerges, this would suggest that the buddy system produces no differential benefit over naturally occurring social adjustment, challenging the assumption that structured interest-based contact is necessary for adaptation. Should SDT need satisfaction increase without corresponding improvements in sociocultural adaptation, this would challenge the proposed mediating role of need fulfilment, suggesting instead that Homophily or contact frequency alone drives adaptation outcomes. Finally, if both groups show equivalent improvement over time, this would indicate that general university adjustment, rather than programme participation specifically, accounts for observed changes, underscoring the key limitation of the quasi-experimental design.

It is further predicted that triad dynamics may introduce variability in outcomes across participants. Drawing on Allport's (1954) conditions for effective intergroup contact, unequal cultural familiarity between Chinese and UK home students may create explicit power dynamics within triads, potentially undermining the equal status conditions necessary for relatedness

fulfilment. These dynamics may not be fully captured by group-level analysis alone, suggesting supplementary qualitative data may be valuable.

Discussion

The present study proposed a longitudinal quasi-experimental validation of the Cross-Cultural Buddy System, examining whether interest-based peer matching, grounded in Homophily (McPherson et al., 2001), Self-Determination Theory (Ryan & Deci, 2000), and Intergroup Contact Theory (Allport, 1954), produces greater improvements in SDT need satisfaction and sociocultural adaptation among first-year Chinese international students in UK universities compared to eligible non-participants. Should the predicted results be supported, the present study would provide preliminary evidence that interest-based peer matching produces meaningful improvements in SDT need satisfaction and sociocultural adaptation among Chinese international students in UK universities. Specifically, significant Group $\times$ Time interactions in BPNS relatedness and competence subscales would validate the theoretical proposition that Homophily driven matching creates the psychological conditions necessary for SDT fulfilment, consistent with McPherson et al., (2001) and Ryan & Deci (2000). Furthermore, significant mediation of sociocultural adaptation outcomes through SDT need satisfaction would support the causal chain proposed in this study, that need fulfilment rather than social exposure alone, is the operative mechanism driving adaptation. These findings would position the Cross-Cultural Buddy System within the Catalyse phase of the DPM (van Zalk & Sheldrick, 2025), providing empirical validation of its core psychological mechanisms and laying the foundation for broader implementation.

Beyond the immediate validation of the buddy system, these findings would carry broader implications for UK university policy and international student support. Universities increasingly

recruit Chinese international students as a significant revenue stream, yet institutional support for their sociocultural integration remains largely transient and high-pressure (Ploner, 2018). A validated, scalable interest-based matching programme offers a low-cost, psychologically grounded alternative to generic welcome events, with potential applicability to other underserved international student populations. Furthermore, the study contributes to the growing body of work applying the Design Psychology Model (van Zalk & Sheldrick, 2025) to real-world intervention design, demonstrating how psychological theory can be systematically embedded into university structures.

Several unintended consequences warrant consideration within the DPM's Monitor stage. First, the sub-dyadic clustering, whereby two members of the triad form a stronger bond, leaving the third feeling peripheral, may paradoxically reinforce social isolation for some participants, undermining the relatedness need the programme aims to fulfil (Allport, 1954). Second, the £3 subsidy, while designed to reduce financial barriers, risks undermining intrinsic motivation over time by shifting participation from autonomous to externally regulated engagement, thereby threatening the autonomy need central to SDT (Ryan & Deci, 2000). Third, UK home students may experience an implicit cultural burden given their familiarity advantage, potentially creating unequal interaction dynamics that contravene the equal-status conditions necessary for effective intergroup contact (Allport, 1954). These dynamics suggest that ongoing qualitative monitoring of triad interaction quality, consistent with DPM's Improvement cycle, would be essential for identifying and addressing unintended harms in real-world deployment.

Several methodological limitations should be acknowledged. First, the quasi-experimental design introduces self-selection bias. Students who voluntarily opt into the buddy system may have higher baseline intercultural motivation and openness to cross-cultural engagement compared to

those who opt out. This pre-existing disposition, rather than programme participation per se, may account for some or all the observed improvements in SDT need satisfaction and sociocultural adaptation, meaning the true effect of the intervention may be overestimated, and causal inference remains limited. Second, reliance on self-report measures introduces the possibility of social desirability bias, particularly if participants infer the study's purpose and respond accordingly. Third, the first academic year may be insufficient to capture the full trajectory of sociocultural adaptation, as meaningful cross-cultural relationships may require longer than one year to develop and stabilise.

Future research should address these limitations by employing random assignment to interest-based versus random matching conditions, thereby isolating the specific contribution of Homophily to adaptation outcomes. A longer follow-up period extending beyond the first academic year would additionally allow examination of whether SDT need satisfaction and sociocultural adaptation gains are sustained over time. Incorporating supplementary qualitative methods, such as post-meetup reflections or semi-structured interviews, would further illuminate the relational dynamics within triads that quantitative measures alone cannot capture, providing a richer understanding of the mechanisms through which the buddy system produces its effects.

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